KeyShield

Proactive Local Protection Against API Key Leakage for AI Developers

Problem Statement

Why It Matters Now

Critical Risks for AI Developers:

- Post-hoc tools (TruffleHog, GitGuardian) detect secrets **after** they are exposed.
- Credential theft tools scrape and misuse keys **within minutes** of exposure.
- **No lightweight local solutions** exist without invasive keyboard logging.

Why Now?

- Explosive LLM adoption → More API keys in use.
- Solo devs & small teams lack dedicated security.
- Open-source culture increases accidental exposure risk.

Solution: KeyShield's Two-Layer Defense

Layer 1: Real-Time File Monitor

- Runs in the background, watching file changes.
- **Instant desktop popup** when LLM API keys are detected.

Layer 2: Git Pre-Commit Hook

- **Blocks commits** containing exposed secrets.
- Prevents accidental uploads to GitHub.

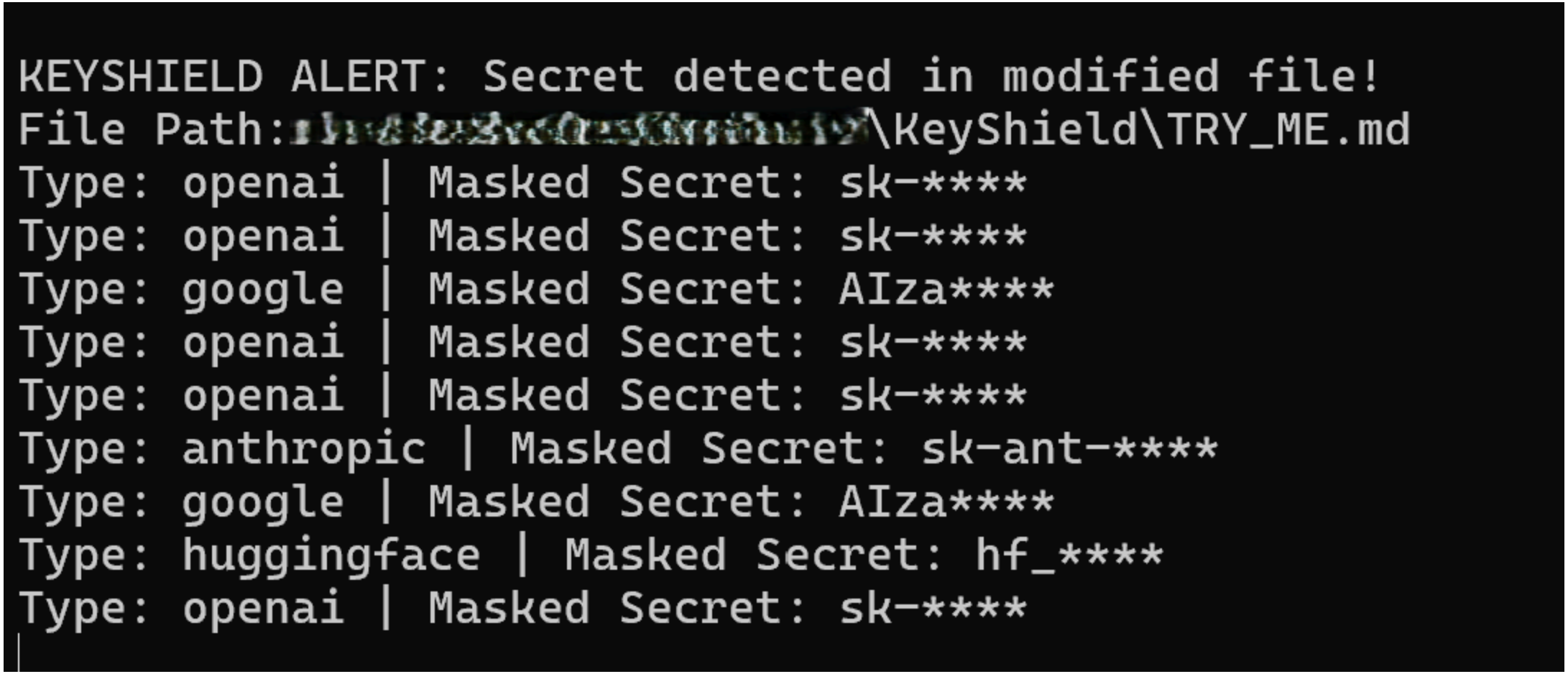
Key Principles:

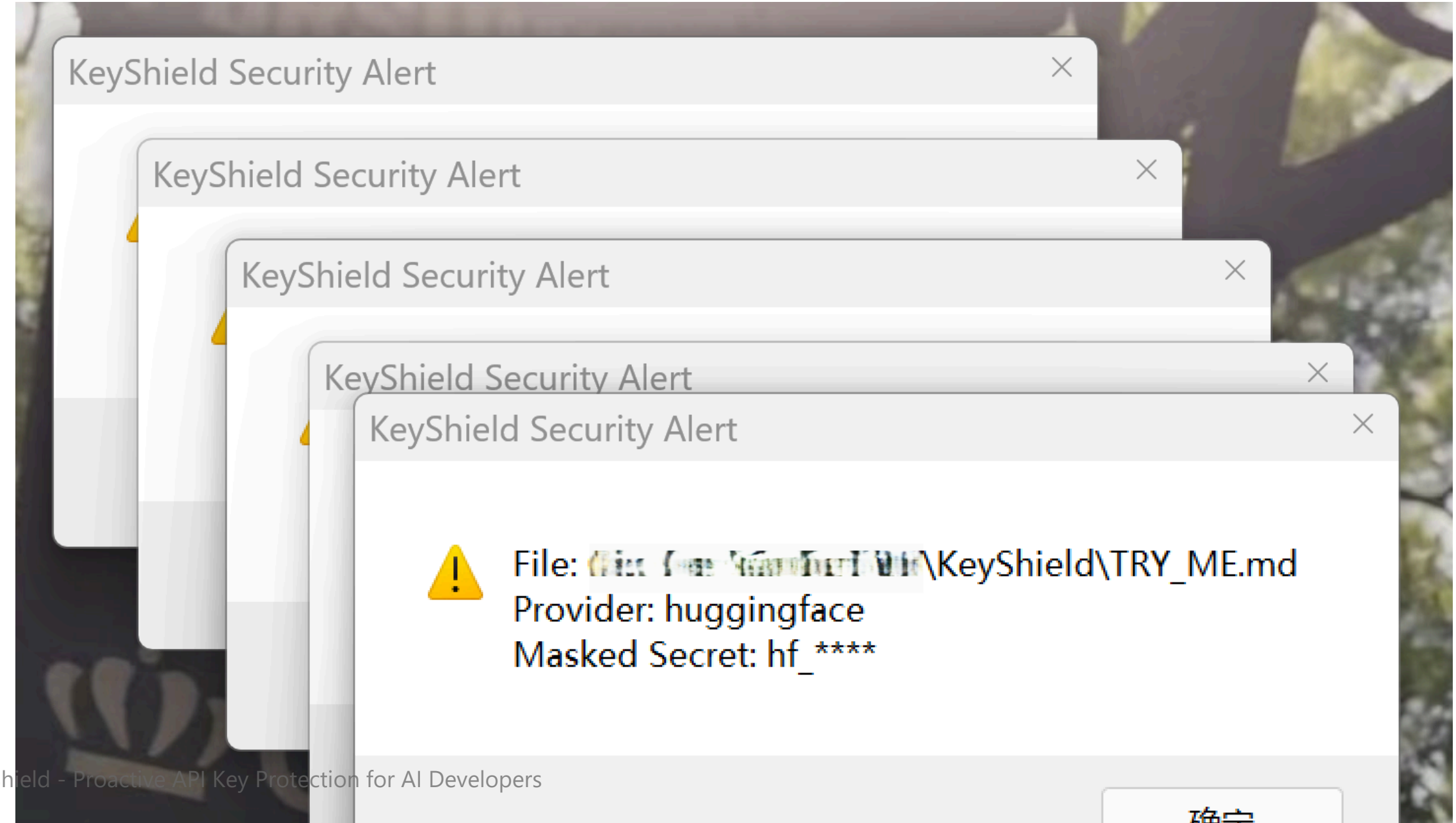
✅ **Zero Persistence** – No database, no login, never stores secrets.

✅ **Privacy-First** – Only scans file content, **never monitors input**.

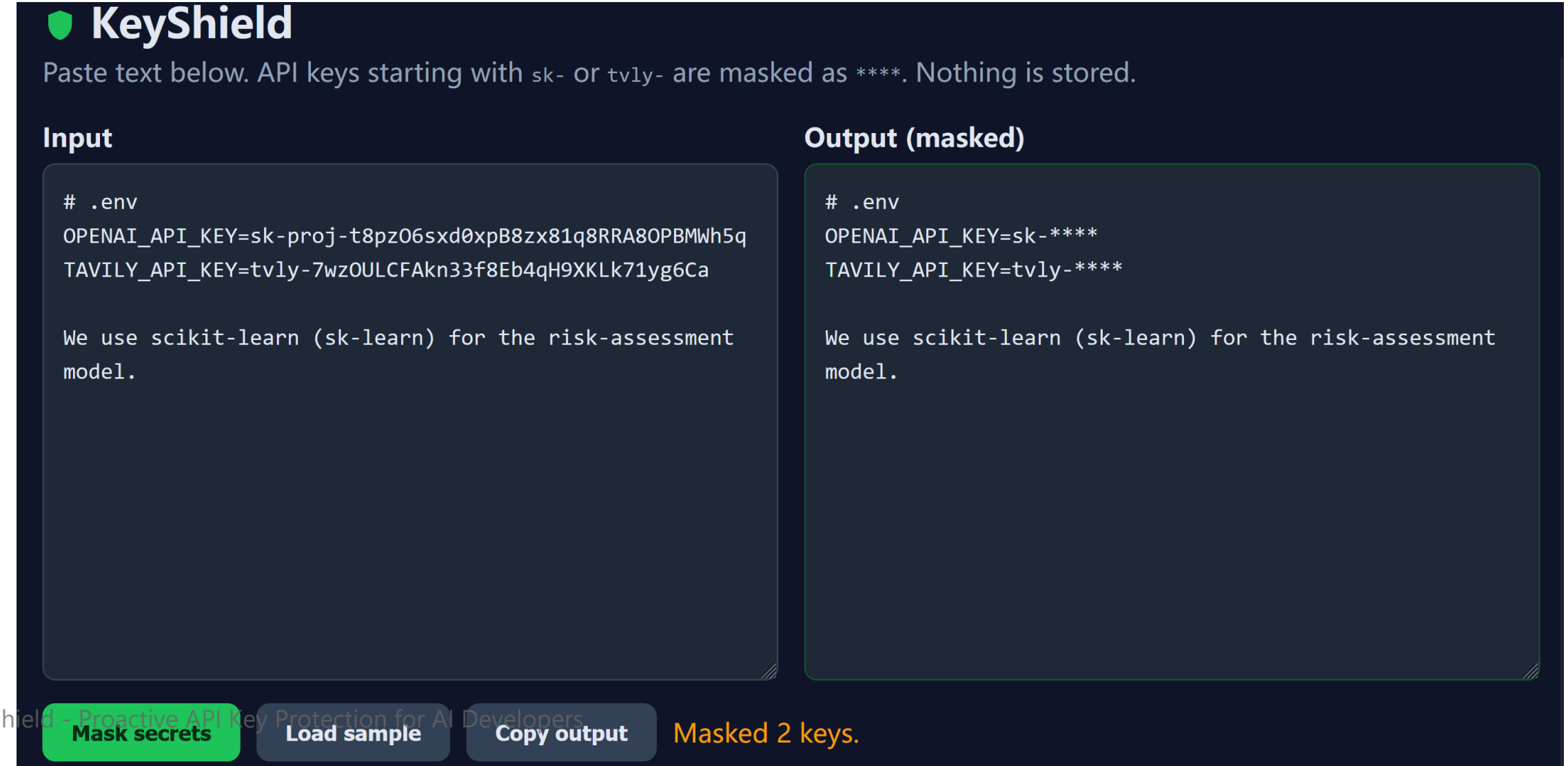
✅ **Lightweight** – Minimal dependencies, instant detection.

Working Prototype: Desktop Alerts




KeyShield Security Alert
KeyShield Security Alert
KeyShield Security Alert
KeyShield Security Alert
KeyShield Security Alert
File: \KeyShield\TRY_ME.md
Provider: huggingface
Masked Secret: hf_****

Working Prototype: Web UI



Market Size & Opportunity

Total Addressable Market (TAM)

- **$2.5B** (Global developer security tools, 2024).
- **15% CAGR** due to API growth and security awareness.

Serviceable Addressable Market (SAM)

- **500K** individual AI developers & small teams globally.
- **200K** developers using 3+ LLM providers.
- **100K** open-source-first users.

Revenue Model

Tier	Features	Price
Free	Basic monitoring (5 providers), community support	Free
Pro	All 10+ providers, priority support, custom patterns, auto-masking	**$9/month**
Enterprise	Team dashboard, SSO, audit logs, Slack/Teams integration	**$49/month**

Additional Revenue:

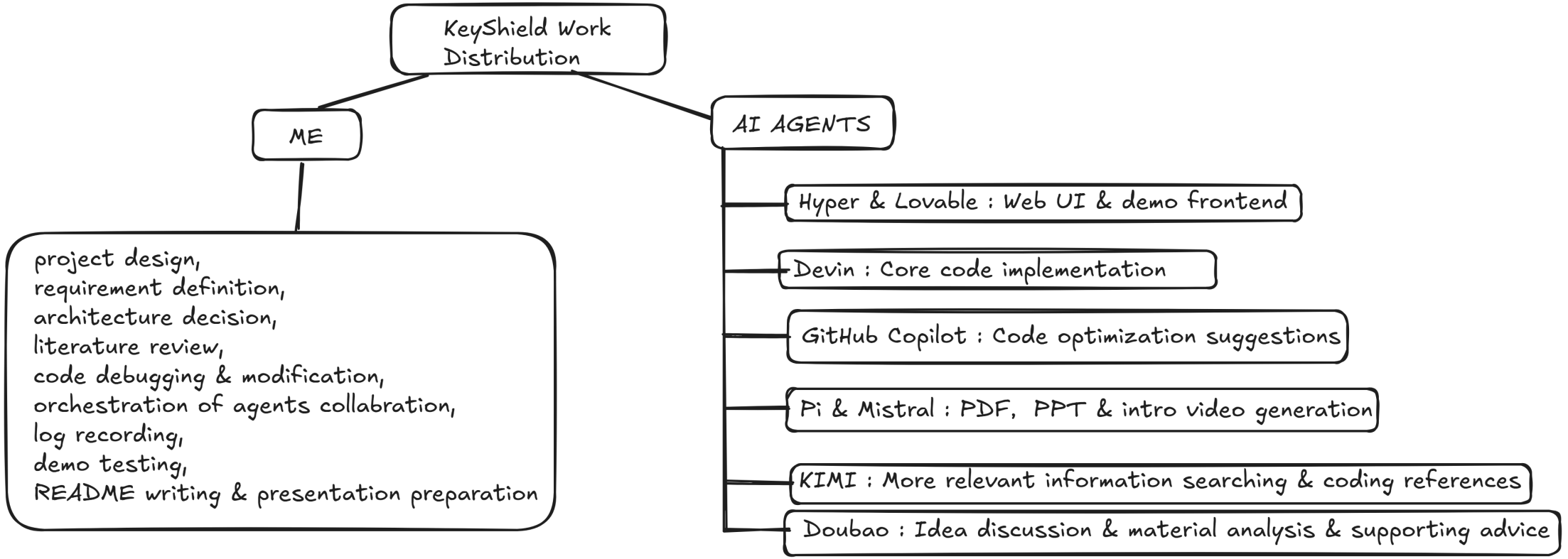
- Custom pattern consulting.
- White-label licensing.
- Training & onboarding.

Competitive Advantage

Feature	TruffleHog	GitGuardian	Gitleaks	KeyShield
Proactive Detection	❌ (Post-commit)	❌ (Post-commit)	❌ (Post-commit)	✅ **Real-time**
Local-Only	❌ (Server-based)	❌ (Cloud)	✅	✅ **Zero cloud**
Desktop Alerts	❌	❌	❌	✅ **Instant popups**
Git Hook Integration	❌	✅	✅	✅ **Auto-blocking**
Privacy-First	✅	❌	✅	✅ **No keyboard logging**

Team

- **Built by**: Solo developer + AI agent team.
- **Work Distribution**:



Roadmap

Phase 1: MVP (Current)

✅ Open-source core with real-time monitoring and git hooks.
✅ Support for 10+ LLM providers.

Phase 2: Community Growth

- Expand pattern library (AWS, Azure, GCP).
- Linux/Mac desktop notifications.
- Contributor onboarding.

Phase 3: Monetization

- Launch **Pro Tier** ($9/month).
- Mobile companion app.

Impact & Growth Potential

Security Impact

- **Goal**: Reduce API key leakage incidents by **80%** for individual developers.
- Eliminate remediation time spent on leaked credentials.

Developer Productivity

- No more panic over accidental commits.
- Seamless workflow with instant feedback.

Open Source & Education

- Foster a **security-first culture** in AI development.
- Raise awareness about secret management best practices.

Key Takeaways

✅ **Proactive protection** – Stops leaks **before** they happen.
✅ **Privacy-first** – No cloud, no logging, no keyboard monitoring.
✅ **Developer-friendly** – Lightweight, easy setup, instant feedback.

Thank You!

Next Steps

- **Try it**: GitHub Repository (Link to repo).
- **Join the community**: Contribute patterns, report issues.
- **Upgrade to Pro**: Unlock premium features.

Contact

- **Project**: KeyShield
- **License**: MIT Open Source
- **Built for**: AI GENESIS Hackathon

Appendix: Quick Start

```
# Clone and install
git clone <repository-url>
cd KeyShield
python -m venv .venv
source .venv/bin/activate  # Linux/Mac
.venv\Scripts\activate     # Windows
pip install -r requirements.txt

# Run real-time monitor
python -m app.monitor

# Install git hook
cp pre_commit_hook.py .git/hooks/pre-commit
chmod +x .git/hooks/pre-commit  # Linux/Mac
```

Appendix: Supported LLM Providers

- OpenAI
- Anthropic
- Google
- Cohere
- Hugging Face
- Mistral
- Together AI
- Perplexity
- Replicate
- Tavily